Problem Set: 4
Computer Networks and the Internet: Network Layer

1. State situations where DHCP is recommended to be used and others where is not recommended to be used.

2. Do routers have IP addresses? If so, how many?

3. The DHCP specifications allow more than one DHCP server to be on the same physical network.
   (a) Why would that be useful?
   (b) What problems does the existence of multiple servers on the same network introduce?
   (c) How would you solve these problems? (You can mention what the standard protocol indicates or your own thinking)

4. List some applications that would benefit from ATMs CBR service model.

5. Suppose datagrams are limited to 1,500 bytes (including header) between source Host A and destination Host B. Assuming a 20-byte IP header, how many datagrams would be required to send an MP3 consisting of 5 million bytes? Explain how you computed your answer.

6. Consider a datagram network using 32-bit host addresses

   Suppose a router has four links, numbered 0 throughout 3, and the packets are to be forwarded to the link interface as follows:

Address Range	LinkInterface
00001110 00000000 00000000 00000000 to 00001110 00111111 11111111 11111111	0
00001110 01000000 00000000 00000000 to 00001110 01000000 11111111 11111111	1
00001110 01000001 00000000 00000000 to 00001110 01111111 11111111 11111111	2
Otherwise	3

   show the forwarding table that has five entries, uses longest prefix matching and forward packets to the correct link interfaces.

Good Luck